

Jinho Cha, Ph.D. | Curriculum Vitae

Faculty, Computer Programming – Gwinnett Technical College

Lawrenceville, Georgia, USA

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ACADEMIC PROFILE

Quantitative business researcher developing data-driven decision systems at the intersection of business analytics, operations and decision sciences, and artificial intelligence.

Research Fields: Business Analytics; Operations & Decision Sciences; AI-Enabled Decision Making; Predictive-Prescriptive Analytics; Revenue Management; Supply Chain & Logistics Analytics; FinTech & Digital Business.

Methodological Expertise: Machine Learning; Statistical Modeling; Stochastic and Robust Optimization; Inverse Optimization; Simulation; Data-Driven Decision Intelligence under Uncertainty.

Managerial Application Domains: Service Operations; Revenue and Capacity Management; Digital Platforms; Healthcare Analytics; Transportation and Logistics; Technology-Enabled Decision Systems.

EDUCATION

Clemson University <i>Ph.D. in Industrial Engineering</i>	Clemson, SC, USA 2015
University of Florida <i>M.S. in Industrial Engineering</i>	Gainesville, FL, USA 2012
Texas A&M University <i>M.S. in Industrial Engineering</i>	College Station, TX, USA 2007
Korea Military Academy <i>B.S. in Electronic Engineering</i>	Seoul, South Korea 1999

ACADEMIC & RESEARCH APPOINTMENTS

Computer Science Division, Gwinnett Technical College <i>Faculty, Computer Programming</i> Conduct AI/data research while teaching programming and computational problem solving.	Lawrenceville, GA, USA 2025–Present
Korea Research Institute for Defense Technology Planning and Advancement (KRIT) <i>Senior Researcher</i> Led and contributed to quantitative R&D spanning analytics, simulation, logistics, technology evaluation, and data-driven decision support.	South Korea 2021–2025
Korea Military Academy <i>Assistant Professor, Department of Mathematics</i> Taught probability, statistics, linear algebra, linear programming, and calculus; conducted quantitative research. Concurrent appointment: Head of Strategic Planning, Army Culture Innovation Center.	Seoul, South Korea 2017–2021
Republic of Korea Army Headquarters and Joint Chiefs of Staff <i>Military Operations Analyst / Operations Research</i> Applied simulation, operations research, and quantitative decision methods to large-scale planning and multinational operational exercises.	South Korea Prior to 2017

Regular.....

All items are regular full-length journal articles. Published and accepted articles are listed together; accepted items are identified explicitly. Jinho Cha's name is shown in bold; * denotes corresponding author. Bibliographic details follow verified journal records where available.

- [J1] Cha, J.* (2026). *Dynamic Inverse Optimization under Drift and Shocks: Theory, Regret Bounds, and Applications*. **IISE Transactions**. Accepted for publication; final publication checks completed. (Sole Author and Corresponding Author)
- [J2] Cha, J.* (2026). *Machine Learning-Driven Overbooking Optimization under Dynamic Cancellation Risk: Evidence from Hotel Revenue Management*. **Computers & Industrial Engineering**, 222, 112341. DOI: [10.1016/j.cie.2026.112341](https://doi.org/10.1016/j.cie.2026.112341).
- [J3] Cha, J., & Lim, S.Y.* (2026). *Bidirectional Calibration-Robust Booking Control under Heterogeneous Cancellation Risk*. **Journal of Industrial and Management Optimization**. Accepted for publication.
- [J4] Shin, H.K., Lee, C., Cha, J., Kil, M., & Choi, J.* (2026). *Adaptive Switching Optimization for Heterogeneous Decision Modules under Operational Uncertainty*. **Journal of Industrial and Management Optimization**, 22(8), 4088–4137. DOI: [10.3934/jimo.2026144](https://doi.org/10.3934/jimo.2026144).
- [J5] Cha, J.*, Pham, L., Minh Ngo Thi, Ho Thi Hien, & Hai Thi Bich Vu (2026). *Distributionally Robust Automation under Overdispersed Count Risk: Ambiguity Sets, Tail Protection, and Oracle-Relative Policy Gap*. **Journal of Industrial and Management Optimization**, 22(8), 4039–4087. DOI: [10.3934/jimo.2026143](https://doi.org/10.3934/jimo.2026143).
- [J6] Le, A.D., Pham, L., Cha, J., Ho, H., Zúñiga, M.A.*, & Mollick, J. (2026). *E-Government Success: The Role of Service Quality, Perceived Value, Trust, and Satisfaction*. **International Journal of Electronic Government Research**, 22(1), 1–29. DOI: [10.4018/IJEGR.411889](https://doi.org/10.4018/IJEGR.411889).
- [J7] Cha, J.*, Han, S.-M., Pham, L., Mollick, J., & Yu, J. (2026). *Optimal Automation under Overdispersed Discrete Risk: Thresholds and Hysteresis in a Negative Binomial Model*. **Journal of Industrial and Management Optimization**, 22(5), 2503–2554. DOI: [10.3934/jimo.2026092](https://doi.org/10.3934/jimo.2026092).
- [J8] Cha, J.*, Yu, J., Cha, E.D., Yoo, E., Geoffrey, C., & Song, H. (2026). *Mechanism Design and Equilibrium Analysis of Smart Contract-Mediated Resource Allocation*. **Journal of Industrial and Management Optimization**, 22(2), 997–1033. DOI: [10.3934/jimo.2026037](https://doi.org/10.3934/jimo.2026037).
- [J9] Cha, J.*, Cha, E.D., Yoo, E., & Song, H. (2025). *Modeling ROI in Chronic Disease Management: A Simulation-Based Framework Integrating Patient Adherence and Policy Timing*. **BMC Public Health**, 25, 4270. DOI: [10.1186/s12889-025-25279-3](https://doi.org/10.1186/s12889-025-25279-3).
- [J10] Ho, H., Han, S.M., Cha, J., & Pham, L.* (2025). *Mobile Banking Customer Satisfaction and Loyalty: The Roles of Technology Readiness*. **Journal of Risk and Financial Management**, 18(7), 403. DOI: [10.3390/jrfm18070403](https://doi.org/10.3390/jrfm18070403).
- [J11] Kim, S.K., Kim, D., Cha, J. (2024). *A Study on Developing ROK Ammunition Container*. **Journal of the Korea Academia-Industrial Cooperation Society**, 25(5), 603–610. DOI: [10.5762/KAIS.2024.25.5.603](https://doi.org/10.5762/KAIS.2024.25.5.603).
- [J12] Lee, H., Cha, J.* (2023). *A Study on the Recognition of Korea Armed Forces Nursing Academy and Military Nursing Officers through Web Crawling and Text Mining*. **Journal of the Korea Academia-Industrial Cooperation Society**, 24(5), 381–388. DOI: [10.5762/KAIS.2023.24.5.381](https://doi.org/10.5762/KAIS.2023.24.5.381).

- [J13] Han, C., Kim, J., **Cha, J.**, Lee, J., Jung, Y., Park, J., Kim, Y.T., Kim, Y.C., Ha, J., Lee, K., Kim, Y.S., & Bang, S. (2022). *A Methodology of AI Learning Model Construction for Intelligent Coastal Surveillance*. **Journal of Internet Computing and Services**, 23(1), 77–86. DOI: [10.7472/jksii.2022.23.1.77](https://doi.org/10.7472/jksii.2022.23.1.77).
- [J14] Kim, S., **Cha, J.**, Lee, J., & Han, C. (2021). *Big Data Analysis to Derive Improvement Direction for Military Mental Force Education: Focusing on the Corpus of Internet News Comments*. **Journal of Spiritual & Mental Force Enhancement**, 65, 3–50. DOI: [10.23054/jsmfe.2021..65.001](https://doi.org/10.23054/jsmfe.2021..65.001).
- [J15] Kim, J., Park, S., **Cha, J.**, & Kim, Y. (2021). *Future Tactical Communication System Development Plan*. **Journal of Convergence for Information Technology**, 11(6), 14–23. DOI: [10.22156/CS4SMB.2021.11.06.014](https://doi.org/10.22156/CS4SMB.2021.11.06.014).
- [J16] Kim, J., Park, S., **Cha, J.**, & Kim, Y. (2021). *Future Tactical Communication System Development Plan through Army TIGER Information Distribution Capability Analysis*. **Journal of Convergence Security**, 21(4), 23–30.
- [J17] Yoon, S.H., **Cha, J.** (2019). *Convolutions and Determination of Number of Random Variables in Truncated Environment*. **Korea Military Academy Journal**.
- [J18] Krenek, R., **Cha, J.**, Cho, B.R., & Sharp, J.L. (2017). *Development of Statistical Convolutions of Truncated Normal and Truncated Skew Normal Distributions with Applications*. **Journal of Statistical Theory and Practice**, 11, 1–25. DOI: [10.1080/15598608.2016.1244022](https://doi.org/10.1080/15598608.2016.1244022).
- [J19] Krenek, R., **Cha, J.**, & Cho, B.R. (2016). *Development of the Convolutions of Truncated Normal Random Variables with Three Different Quality Characteristics in Engineering Applications*. **Computers & Industrial Engineering**, 94, 125–137. DOI: [10.1016/j.cie.2015.12.014](https://doi.org/10.1016/j.cie.2015.12.014).
- [J20] Krenek, R., **Cha, J.**, & Cho, B.R. (2015). *Zippping and Re-Zipping Methods to Improve the Precision and Accuracy of Manufacturing Processes*. **International Journal of Experimental Design and Process Optimisation**, 4(3/4), 256–289. DOI: [10.1504/IJEDPO.2015.072828](https://doi.org/10.1504/IJEDPO.2015.072828).
- [J21] **Cha, J.**, & Cho, B.R. (2015). *Classical Statistical Inference Extended to Truncated Populations for Continuous Process Improvement: Test Statistics, P-values, and Confidence Intervals*. **Quality and Reliability Engineering International**, 31(8), 1807–1824. DOI: [10.1002/qre.1719](https://doi.org/10.1002/qre.1719).
- [J22] **Cha, J.**, Cho, B.R., & Sharp, J.L. (2013). *Rethinking the Truncated Normal Distribution*. **International Journal of Experimental Design and Process Optimisation**, 3(4), 327–363. DOI: [10.1504/IJEDPO.2013.059667](https://doi.org/10.1504/IJEDPO.2013.059667).

DOCTORAL DISSERTATION.....

- **Cha, J.** (2015). *Re-Establishing the Theoretical Foundations of a Truncated Normal Distribution: Standardization, Statistical Inference, and Convolution*. Clemson University, Department of Industrial Engineering.

MANUSCRIPTS UNDER REVIEW

All manuscripts in this section are currently under review.

- [1] **Cha, J.**, et al. *Generalization, Minimax Limits, and Regret in Outcome-Only Strategic-Response Learning under Misspecification*. **Journal of Machine Learning Research**. Under review. (First Author)
- [2] **Cha, J.*** *Difficulty-Controlled Benchmarking for Inverse Discrete Optimization*. **Computers & Operations Research**, Elsevier. Under review. (Sole Author and Corresponding Author)
- [3] **Cha, J.*** *Revealed-Objective Conformance Certification for Black-Box Artificial Intelligence Systems*. **IEEE Transactions on Artificial Intelligence**. Under review. (Sole Author and Corresponding Author)

- [4] **Cha, J.***, et al. *Geographic Disparities in Hospice Quality and Family Caregiver Experience: Ownership, Social Vulnerability, and County Operating Context*. **Health Care Management Science**, Springer Nature. Under review. (First and Corresponding Author)
- [5] **Cha, J.**, et al. *Optimal Procurement and Technology Adoption under Carbon Regulation: A Dual-Threshold Dynamic Framework*. **Annals of Operations Research**, Springer Nature. Under review. (First Author)
- [6] **Cha, J.*** *Bilevel Credit Screening with Strategic Feature Adjustment and Verification*. **Operations Research Letters**, Elsevier. Under review. (Sole Author and Corresponding Author)
- [7] **Cha, J.**, Pham, L., & Vo, T.L.H.* *A Finite Breakpoint Rule for Carbon-Adjusted Last-Mile EV Capacity Allocation*. **Operations Research Letters**, Elsevier. Under review. (First Author)
- [8] **Cha, J.***, Cho, J., Lee, J., & Shin, J. *Uncertainty-Aware Knowledge Discovery from Peer-Review Response Traces: Taxonomy-Free Structure Discovery, Difficulty-Weighted Recovery, and Evidence Certification*. **Knowledge and Information Systems**, Springer Nature. Under review. (First and Corresponding Author)
- [9] Shin, H.K., **Cha, J.**, & Yang, Y.-G. *Architectural Stress Testing for Generative AI in C2: A Decision-Centric Evaluation of Execution Binding*. **IEEE Systems Journal**. Under review. (Second Author)
- [10] Shin, H.K., Lee, S., Choi, J., & **Cha, J.*** *Projected-Future Evidence Objects for Traceable Dispatch in AI-Enabled Systems of Systems*. **IEEE Systems Journal**. Under review. (*Corresponding and Submitting Author)
- [11] **Cha, J.*** *Digital SAF Book-and-Claim Markets: Verification Investment and Incentive Design under Imperfect Authentication*. **Industrial Management & Data Systems**, Emerald Publishing. Under review. (Sole Author and Corresponding Author)
- [12] **Cha, J.**, Lim, S.Y., & Park, H.* *When Does Hotel Cancellation Prediction Create Economic Value? Calibration, Decision Boundaries, and Booking Control*. **Tourism Economics**, SAGE. Under review. (First Author)
- [13] **Cha, J.***, Cha, E.D., Yu, J., Roh, M., & Park, H. *LLM-Assisted Decision Summaries for End-of-Life Care Planning in the ICU: A Documentation-Centered Framework Using MIMIC-IV*. **BMC Medical Informatics and Decision Making**, Springer Nature. Under review. (First and Corresponding Author)
- [14] Shin, H.K., **Cha, J.**, et al. *Non-Bypassable Decision Execution Orchestration for Authority-Aligned Autonomous Systems*. **Systems Engineering**, Wiley. Under review. (Second Author)
- [15] Vu, L.T., **Cha, J.***, Nguyen, T.-T., & Pham, L. *Customer Complaint Analytics for Service Recovery in Digital Financial Services: Machine Learning and Capacity-Aware Prioritization*. **Future Business Journal**, Springer Nature. Under review. (Second and Corresponding Author)
- [16] **Cha, J.**, Pham, L., Cho, J., & Vo, T.L.H.* *Pricing Verification Risk in Sustainable Aviation Fuel Book-and-Claim Markets: Guarantee Design*. **Green Finance**, AIMS Press. Under review. (First Author)
- [17] **Cha, J.***, et al. *Data-Driven Optimization for Airline Service Quality Improvement under Delay and Cancellation Risk*. **Journal of Industrial and Management Optimization**. Under review. (First and Corresponding Author)
- [18] **Cha, J.**, & Kwon, E.-Y.* *From Health-AI Risk Scores to Budget-Feasible Decisions: Predictive Optimization for Public Health Resource Allocation*. **Journal of Industrial and Management Optimization**. Under review. (First Author)

- [19] Kim, Y., Lee, C., Hwang, H., **Cha, J.**, & Park, H.* *Stochastic Production-Readiness Design and Order-Acceptance Scheduling in High-Mix Low-Volume Manufacturing*. **Journal of Industrial and Management Optimization**. Under review.
- [20] Lam, V.T., Pham, L., & **Cha, J.*** *When Should a Financial Forecasting Model Be Renewed? Multimetric Degradation Monitoring and Candidate-Specific Valuation under Market Drift*. **Data Science in Finance and Economics**, AIMS Press. Under review. (*Corresponding Author)
- [21] Kang, J., & **Cha, J.*** *AI-Aware Cyber Response Prioritization under Accelerated Vulnerability Threats: A Risk-Informed Scheduling Framework*. **KSII Transactions on Internet and Information Systems**. Under review. (*Corresponding Author)
- [22] Shin, H.K., **Cha, J.**, et al. *Foundation-Model-Based Decision-Intent Orchestration for Decision-Centric Command and Control Systems*. **Journal of Defense Modeling and Simulation**, SAGE. Under review. (Second Author)
- [23] Hwang, H., & **Cha, J.*** *Integrated Predictive Maintenance and Production Scheduling under Limited Maintenance Resources*. **Industrial Engineering & Management Systems**. Under review. (*Corresponding Author)

TEACHING PROFILE

TEACHING INTERESTS.....

Business Analytics; Operations Management and Decision Models; Applied Statistics for Business; Predictive–Prescriptive Analytics; Data Analytics and Programming for Business; AI for Managerial Decision Making; Optimization and Quantitative Methods.

TEACHING EXPERIENCE.....

Computer Science Division, Gwinnett Technical College

Lawrenceville, GA, USA

Faculty, Computer Programming

2025–Present

Courses taught: CIST 2371 – Java Programming I; CIST 1305 – Program Design and Development.

Terms taught: Summer 2025; Fall 2025; Spring 2026; Summer 2026; Fall 2026.

Department of Mathematics, Korea Military Academy

Seoul, South Korea

Assistant Professor

2017–2021

Courses taught: Calculus I; Calculus II; Advanced Calculus (Multivariate); Differential Equations; Linear Algebra; Probability and Statistics; Linear Programming; Theory of Probability.

Student evaluation ratings exceeded 4.5/5.0 across listed courses.

Department of Industrial Engineering, Clemson University

Clemson, SC, USA

Teaching Assistant

2013–2015

Courses supported: IE 4610 – Quality Engineering; IE 4620/6620 – Six Sigma Quality.

Clemson University Center for Workforce Development

Clemson, SC, USA

Research Assistant

2012–2013

Contributed to a Six Sigma-based operational improvement project.

RESEARCH GRANTS & SPONSORED PROJECTS

Selected funded projects; roles, sponsors, award amounts, and periods are shown consistently.

Co-PI • AI-Driven Full Stack Development Toolkit for Emerging Technicians

\$10,000

NITIC Mini-Grant Program, Gwinnett Technical College

2025–2026

Co-led AI and mathematical-modeling components, including algorithmic logic, optimization, and instructional design for AI-assisted full-stack development.

Co-PI • Smart Grader: An AI-Driven Code Assessment and Feedback System with Blockchain Logging	\$10,000
<i>NITIC Mini-Grant Program, Gwinnett Technical College</i>	2025–2026
Co-led AI-driven code assessment and evaluation, real-time feedback design, learning analytics, and blockchain-style logging for technician-level programming education.	
Project Manager • Development of Korea Military Fiber Ammunition Container	\$1,000,000
<i>Ministry of National Defense, Republic of Korea</i>	2021–2024
Led a national R&D program from problem definition and analytical design through validation and implementation.	
Co-PI • Development of Intelligent Coastal Surveillance System	\$750,000
<i>Korea Army Training and Doctrine Command</i>	2020–2022
Co-led a surveillance-system R&D project integrating sensing, analytics, and decision-support capabilities.	
PI • Cost and Operational Analysis of Army Scientific Training Systems	\$100,000
<i>Republic of Korea Army</i>	2019–2020
Led quantitative cost and operational analysis for scientific training-system decisions.	
Co-PI • Cost Analysis of Armored Vehicle Development	\$200,000
<i>Defense Agency for Technology and Quality</i>	2018–2019
Conducted cost-analysis research supporting defense acquisition and technology decisions.	
PI • Study on Officer Human Resource Acquisition and Development	\$200,000
<i>Republic of Korea Army</i>	2018–2019
Led data-informed analysis of officer acquisition and human-resource development.	
Co-Project Manager • Maintenance Project for Ground Warfare Model at Corps Level and Above	\$450,000
<i>Republic of Korea Army</i>	2007–2010
Co-managed maintenance and analytical support for a large-scale ground-warfare simulation model.	

HONORS & AWARDS

Presidential Commendation for Contributions to National Defense Capability Enhancement	Republic of Korea, 2025
Ministerial Commendation for Enhancements to Logistics Requirements System	Ministry of National Defense, 2023
Army Chief of Staff Commendation for Contributions to Army Innovation	2019
Superintendent’s Citation, Korea Military Academy	2018
Chairman of the Joint Chiefs of Staff Commendation	2017

CONFERENCE PRESENTATIONS & INVITED TALKS

- [1] **Cha, J.** (2026, May 15). *A Study on Educational Innovation at the Korea Military Academy Using Artificial Intelligence*. National Defense Innovation Seminar, Korea Military Academy, Seoul, South Korea.
- [2] **Cha, J.** (2025). *Smart Contract Adoption Decisions under Discrete Overdispersed Demand: A Negative Binomial Optimization Perspective*. INFORMS Annual Meeting, Atlanta, GA, USA, October 23, 2025.
- [3] **Cha, J.** (2024, May; 2024, November). *Development Measures for Combatant Supporting Systems using AI for Military*. Invited Lecture, Military AI Requirement Planning and Project Management Program, Sungkyunkwan University, Pangyo, South Korea.
- [4] **Cha, J.** (2023, August). *Military R&D Development Direction of the Korean Armed Forces*. Invited Lecture, Korea Military Academy, Seoul, South Korea.
- [5] **Cha, J.** (2023, May; 2022, May; 2022, November). *Strategic Applications of Artificial Intelligence in*

Combat Support and Defense Systems. Invited Lecture, Sungkyunkwan University, Pangyo, South Korea.

- [6] **Cha, J.**, Han, C.H. (2019). *Analysis of Military Culture in the Army Using Big Data Techniques*. 12th International South Korea Army Modeling & Simulation Education Conference, Daejeon, South Korea, November 19–20, 2019.
- [7] **Cha, J.**, Krenek, R., Cho, B.R. (2017). *Zippping and Re-zipping Methods to Improve the Precision and Accuracy of Manufacturing Processes*. Industrial Engineering Research Symposium, Clemson University, Clemson, SC, May 2017.
- [8] **Cha, J.**, Kim, Y.H. (2015). *A Study on the Federation Security System for the War Game Models for the U.S. and R.O.K Combined Forces Exercise*. 9th International Army Modeling & Simulation Education Conference, Daejeon, South Korea, November 8–9, 2015.
- [9] **Cha, J.**, Krenek, R., Cho, B.R., Sharp, J.L. (2015). *Development of Statistical Inference using Truncated Statistics with Military Applications*. IIE Annual Conference and Expo, Nashville, TN, May 2015.
- [10] **Cha, J.** (2009). *Increasing Interoperation with the US Army's Ground War Game Model*. Ground Weapon Conference, Daejeon, South Korea, June 2009.
- [11] **Cha, J.** (2008). *Performing Interoperation Analyses of Artillery, Aviation, and Air Defense Functions in Ground War-Game Models between U.S. and R.O.K Combined Exercise*. Military Operations Research Conference, Seoul, South Korea, May 2008.

PROFESSIONAL SERVICE

JOURNAL PEER REVIEW.....

- Ad Hoc Reviewer, **ACM Transactions on Knowledge Discovery from Data (TKDD)**, 2026.
- Ad Hoc Reviewer, **PLOS ONE**, 2026.
- Reviewer, MDPI journals: **Sustainability**; **Applied Sciences**; **Mathematics**; **Artificial Intelligence**; **Processes**; and **Electronics**.
- Additional ad hoc peer-review service in operations research, artificial intelligence, analytics, and decision-support applications.

PROFESSIONAL MEMBERSHIP

- **Member**, INFORMS (Institute for Operations Research and the Management Sciences).

PROFESSIONAL & COMMUNITY AFFILIATIONS

- **Vice President**, The Korean Veterans Association, Southern Regional Chapter USA.
- **Member**, Korea Defense Veterans Association (KDVA).

GOVERNMENT & TECHNICAL REPORTS

- **Cha, J.** (Co-Author). *2025–2034 National Defense Force Support System Requirements Plan (Public Edition)*. Ministry of National Defense and Korea Research Institute for Defense Technology Planning and Advancement (KRIT), 2024.

Curriculum vitae audited and updated September 7, 2026.